

All-degree instability of power maps for small fractional order

Automated mathematics run `fa_banach_001`

11 August 2026

Status. Candidate partial result, likely valid. The theorem below is complete, but it does not settle existence of an arbitrary degree-one minimizer.

Abstract

For the conformally invariant $W^{s,1/s}$ energy of circle maps, a recent preprint proves that the identity is unstable when $s < 1/8$ and remarks that the same seems true for every power map, while explicitly describing the supporting computation as AI-generated and unverified. We prove that all-degree statement. On the mode $\cos(3|d|\theta)$, the second variation at $z \mapsto z^d$ is exactly $|d|$ times the degree-one second variation. The reduction follows from a cosecant root-sum identity, and a beta-integral calculation gives the sign factor $8 - 1/s$.

1 Source question and immediate context

Mazowiecka and Schikorra ask about existence of minimizing $W^{s,n/s}(\mathbb{S}^n, \mathbb{S}^n)$ maps of degree one [1, p. 6]. The source passage is reproduced below.

us also mention the flow-technique developed for harmonic maps by Struwe [108] which he used to show the existence of nontrivial minimal harmonic maps (cf. Corollary 7.2). For results concerning 1-harmonic maps we refer to [42]. We also refer to [2].

A brief history of fractional harmonic maps. The theory of fractional harmonic maps, i.e., critical points and minimizers of the energy in (1.2), can be traced back to the 1930s when Douglas [32] used them (implicitly) to solve the Plateau problem and win the Fields price. Analytically they were introduced in the pioneering work by Da Lio and Rivière [28, 27] who coined the notion of fractional harmonic maps and developed the regularity theory for critical (i.e., not necessarily minimizing) $W^{\frac{1}{2},2}$ -harmonic maps on lines into manifolds. This regularity theory for critical points was extended to various variations of the energy functional [94, 24, 29, 97, 98, 30, 71] — in particular the notion of (critical) $W^{s,p}$ -harmonic maps and their regularity theory into spheres was introduced in [96] (see also [72]). The question of existence of minimizing $W^{s,\frac{n}{s}}(\mathbb{S}^n, \mathbb{S}^n)$ -maps of degree one was raised earlier, see Mironescu [77].

In dimension one, Martino–Mazowiecka–Schikorra prove that the identity is not minimizing when $s < 1/8$ [2, Theorem 1.1]. Their footnote says that every z^d appears to lose positive definiteness in the same regime, but labels the trigonometric computation “AI generated, unverified.”

one maps. This is challenging, since the usual methods to show that Möbius transforms minimize degree one maps of an energy — showing that minimizers are conformal maps either by an harmonic extension argument [14, 10], Fourier analysis [15, 1, 4], or realization of sharp geometric inequalities [9] — are unavailable for $W^{s, \frac{1}{s}}$ unless $s = \frac{1}{2}$. Instead our argument builds on what amounts to a precise Morse index analysis combined with the stability analysis *in* s initiated in [13].

It is natural to hope that this holds for any $s \in (0, 1)$, and that $u = \text{id} : \mathbb{S}^1 \rightarrow \mathbb{S}^1$ is a minimizing degree 1 map. Quite surprisingly, but actually not that difficult to prove, the latter is actually false for $s < \frac{1}{8}$, indeed we have our first result

Theorem 1.1. *For any $s \in (0, 1/8)$ the identity map $\text{id} : \mathbb{S}^1 \rightarrow \mathbb{S}^1$ is not a minimizer of E_s among degree 1 maps from $\mathbb{S}^1 \rightarrow \mathbb{S}^1$, i.e. we have $\#_s 1 < E_s(\text{id})$. Indeed, it is not even stable in the $W^{s, \frac{1}{s}}$ -topology.*

Theorem 1.1 follows from a precise computation of the second variation of E_s at the identity, and the observation that it simply fails to be nonnegative¹ definite when $s < \frac{1}{8}$.

¹We note in passing that it seems that this is the case for all degrees $z^n : \mathbb{S}^1 \rightarrow \mathbb{S}^1$, according to (AI generated, unverified, but if need be easily checkable trig integrals) computations, all of them loose positive definiteness at $s < \frac{1}{8}$, and the same seems to happen for slightly different thresholds in higher dimensions for $W^{s, \frac{d}{s}}(\mathbb{S}^d, \mathbb{S}^d)$

The theorem below supplies a proof of that all-degree statement. It is substantial progress around the source question, but does not prove or disprove existence of a different degree-one minimizer.

2 Definitions and statement

For $s \in (0, 1)$ put $p = 1/s$ and define, for $u : \mathbb{S}^1 \rightarrow \mathbb{S}^1$,

$$E_s(u) = \int_0^{2\pi} \int_0^{2\pi} \frac{|u(e^{i\theta}) - u(e^{i\sigma})|^p}{|e^{i\theta} - e^{i\sigma}|^2} d\theta d\sigma. \quad (1)$$

For $d \in \mathbb{Z} \setminus \{0\}$ let $u_d(e^{i\theta}) = e^{id\theta}$ and write $q = |d|$. Consider the smooth, degree- d path

$$u_{d,\varepsilon}(e^{i\theta}) = \exp(i(d\theta + \varepsilon \cos(3q\theta))). \quad (2)$$

Theorem 1 (Exact all-degree negative mode). *Let $s \in (0, 1/8)$, $p = 1/s$, and $d \neq 0$. For the path (2),*

$$\frac{1}{p} \frac{d^2}{d\varepsilon^2} E_s(u_{d,\varepsilon}) \Big|_{\varepsilon=0} = q(8-p)2^{p+1}\pi^{3/2} \frac{\Gamma((p+1)/2)}{\Gamma(3+p/2)} < 0. \quad (3)$$

Consequently u_d is not a local minimizer of E_s within its degree class, even against smooth perturbations. In particular,

$$\inf_{\deg u=d} E_s(u) < E_s(u_d) = qE_s(u_1).$$

3 Proof intuition

The energy is translation invariant in the angular variables, so its second variation at a power map is a Fourier multiplier. The target angle winds q times while the domain angle winds once; this

changes the target factors to $\sin(qt/2)$ but leaves a domain denominator $\sin^2(t/2)$. Choosing the frequency $3q$ synchronizes the perturbation with every target winding. Partitioning the t -circle into q arcs then turns all target factors into their degree-one versions. The only apparent obstruction is the sum of the q domain denominators, and the cosecant root-sum identity collapses it exactly, leaving a factor q . The remaining degree-one beta integral has sign $8 - p$.

4 Proof

We first compute the quadratic form at an arbitrary power map. For a smooth real 2π -periodic function η , set

$$v_\varepsilon(\theta) = e^{i(d\theta + \varepsilon\eta(\theta))}.$$

Write $a = e^{id\theta}$, $b = e^{id\sigma}$, $t = \theta - \sigma$, and $\Delta = dt$. At $\varepsilon = 0$, if $D = a - b$, $V = \partial_\varepsilon(v_\varepsilon(\theta) - v_\varepsilon(\sigma))$, and $W = \partial_\varepsilon^2(\dots)$, then

$$V = \eta(\theta)a^\perp - \eta(\sigma)b^\perp, \quad W = -\eta(\theta)^2a + \eta(\sigma)^2b.$$

Elementary scalar-product identities give

$$\begin{aligned} \langle D, V \rangle &= \sin \Delta [\eta(\theta) - \eta(\sigma)], \\ |V|^2 + \langle D, W \rangle &= \cos \Delta [\eta(\theta) - \eta(\sigma)]^2. \end{aligned}$$

The first variation vanishes: after writing $\theta = \sigma + t$, its inner integral contains $\int_0^{2\pi} (\eta(\sigma+t) - \eta(\sigma)) d\sigma = 0$.

Using

$$\frac{1}{p} \frac{d^2}{d\varepsilon^2} |D_\varepsilon|^p = (p-2)|D|^{p-4} \langle D, V \rangle^2 + |D|^{p-2} (|V|^2 + \langle D, W \rangle),$$

and $|D| = 2|\sin(\Delta/2)|$, we obtain

$$\mathcal{Q}_{p,d}(\eta) := \frac{1}{p} \frac{d^2}{d\varepsilon^2} E_s(v_\varepsilon) \Big|_0 = \int_0^{2\pi} \int_0^{2\pi} K_{p,d}(\theta - \sigma) [\eta(\theta) - \eta(\sigma)]^2 d\theta d\sigma, \quad (4)$$

where

$$K_{p,d}(t) = \frac{|2\sin(dt/2)|^{p-2}}{4\sin^2(t/2)} (p\cos^2(dt/2) - 1). \quad (5)$$

For $p > 8$ all differentiations and integrals are justified directly by dominated convergence; the apparent singularities in (5) are removable after multiplication by the phase difference.

We now record the exact folding identity.

Lemma 1 (Cosecant root sum). *For every integer $q \geq 1$ and $x \notin 2\pi\mathbb{Z}$,*

$$\sum_{j=0}^{q-1} \csc^2\left(\frac{x + 2\pi j}{2q}\right) = q^2 \csc^2(x/2). \quad (6)$$

Proof. The factorization of $\sin(qz)$, or logarithmic differentiation of it, gives

$$\sum_{j=0}^{q-1} \cot\left(z + \frac{\pi j}{q}\right) = q \cot(qz).$$

Differentiate and put $z = x/(2q)$. □

Take $\eta(\theta) = \cos(3q\theta)$. Integrating first in σ in (4) yields

$$\mathcal{Q}_{p,d}(\eta) = 2\pi \int_0^{2\pi} K_{p,d}(t)(1 - \cos(3qt)) dt. \quad (7)$$

The kernel is unchanged when d is replaced by $q = |d|$. Partition the integral into q consecutive intervals and on the j th interval put $t = (x + 2\pi j)/q$. All numerator terms become their degree-one versions. Using Lemma 1,

$$\int_0^{2\pi} K_{p,d}(t)(1 - \cos(3qt)) dt = q \int_0^{2\pi} K_{p,1}(x)(1 - \cos(3x)) dx. \quad (8)$$

Therefore

$$\mathcal{Q}_{p,d}(\cos(3q\cdot)) = q \mathcal{Q}_{p,1}(\cos(3\cdot)). \quad (9)$$

It remains to evaluate the degree-one form. Setting $x = 2r$ in (7) and using $\sin(3r) = \sin r(4\cos^2 r - 1)$ gives

$$\mathcal{Q}_{p,1}(\cos(3\cdot)) = 2^{p-1}\pi \int_0^\pi \sin^{p-2} r (p\cos^2 r - 1)(4\cos^2 r - 1)^2 dr. \quad (10)$$

Let

$$A_k = \int_0^\pi \sin^{p-2} r \cos^{2k} r dr = B\left(\frac{p-1}{2}, k + \frac{1}{2}\right).$$

Relative to A_0 ,

$$\frac{A_1}{A_0} = \frac{1}{p}, \quad \frac{A_2}{A_0} = \frac{3}{p(p+2)}, \quad \frac{A_3}{A_0} = \frac{15}{p(p+2)(p+4)}.$$

Expanding the polynomial in (10), the bracketed integral is

$$16pA_3 - (8p+16)A_2 + (p+8)A_1 - A_0 = A_0 \frac{16(p-1)(8-p)}{p(p+2)(p+4)}.$$

Since $A_0 = \sqrt{\pi} \Gamma((p-1)/2)/\Gamma(p/2)$, the gamma recurrence yields

$$\mathcal{Q}_{p,1}(\cos(3\cdot)) = (8-p)2^{p+1}\pi^{3/2} \frac{\Gamma((p+1)/2)}{\Gamma(3+p/2)}. \quad (11)$$

Combining (9) and (11) proves (3). It is strictly negative for $p > 8$.

Finally, the path (2) has degree d for every ε and converges smoothly, hence in $W^{s,p}$, to u_d . Taylor expansion and the vanishing first variation imply $E_s(u_{d,\varepsilon}) < E_s(u_d)$ for all sufficiently small nonzero ε . The identity $E_s(u_d) = qE_s(u_1)$ follows from the same partition and Lemma 1, now applied directly to (1). This completes the proof of Theorem 1.

5 Verification and limitations

The included script `code/verify_multiplier.py` evaluates (7) with 50-digit quadrature for $p \in \{9, 10, 12\}$ and $q \in \{1, 2, 3, 5\}$. It reproduces q times (11) to roughly 10^{-47} or better. This is only a regression check; the proof is the exact argument above.

This theorem does not settle the source existence question. In particular, when $s < 1/8$ a non-power degree-one map might still attain the infimum. The endpoint $s = 1/8$ is degenerate for the selected mode, and neither other maps nor higher-dimensional analogues are addressed.

A bounded novelty check on 11 August 2026 searched the run indexes and the official arXiv API using the exact titles and combinations of “power maps,” “fractional harmonic,” “second variation,” “degree one,” “ W^s ,” and “positive definiteness.” The exact degree-one query returned only arXiv:2606.15644v1, whose footnote explicitly says the all-degree computation is unverified; the closest all-degree and second-variation queries returned no records. No rigorous prior proof was found. The novelty assessment is plausible, not an exhaustive certification.

Human-review recommendation. Check equations (5), (8), and the beta-integral normalization. These are the only nontrivial algebraic transitions.

References

- [1] K. Mazowiecka and A. Schikorra, *Minimal $W^{s,n/s}$ -harmonic maps in homotopy classes*, J. Lond. Math. Soc. 108 (2023), 742–836; arXiv:2006.07138.
- [2] D. Martino, K. Mazowiecka, and A. Schikorra, *Minimizing and non-minimizing degree one $W^{s,1/s}$ -harmonic maps between spheres*, arXiv:2606.15644v1 (2026).